

James Sungarda

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Education

The University of Hong Kong

BACHELOR OF ENGINEERING (COMPUTER SCIENCE)

Hong Kong SAR

Sep. 2024 - Jun. 2028 (expected)

- Cumulative GPA: 3.41/4.30.

Skills

Programming Languages C++, Python, Javascript/Typescript, HTML/CSS, SQL, Bash, Lua, TeX
Frameworks Next.js, React, Django, Flask, Express, Matplotlib, Pandas, NumPy, PyTorch
Languages English (Native), Mandarin (Fluent), Cantonese (Basic), Indonesian (Native)

Experience

Tam Wing Fan Innovation Wing

RESEARCH ASSISTANT

Hong Kong SAR

Jun. 2021 - Present

- Published and presented a research paper on Generative AI as an educational tool as first author, accepted at **IEEE International Conference on Big Data 2025**, a top-tier academic conference in AI and Data Science with an acceptance rate of **18%**.
- Engineered and deployed a production-grade pedagogical tool supporting **300+ active users** across the entire HKU Department of Social Work and Social Administration; maintained **99.9%** uptime during peak curriculum integration periods.
- Architected and developed a high-traffic full-stack infrastructure using **Next.js, MongoDB, and TailwindCSS**, optimizing database queries to handle **500+ monthly sessions**.
- Developed scalable **RESTful API** endpoints with **Flask** to facilitate real-time LLM inferences, achieving an average latency of **< 200ms** and successfully processing **6,000+** requests without service degradation.

PT. Moxcal Teknologi

SOFTWARE ENGINEER

Jakarta, Indonesia

May 2024 - Aug. 2024

- Developed and designed a web (HTML, Bootstrap, and JavaScript) and WhatsApp based **AI chatbot** and **admin dashboard** to automate customer support.
- Built a Natural Language Processing (NLP) model to accurately **classify user intent** and **extract key entities** from user messages.
- Enabled the chatbot to communicate through both text and natural-sounding speech through **voice recognition** and **text-to-speech**.

Publications

GenAI for Social Work Field Education: Client Simulation with Real-Time

Feedback (<https://arxiv.org/abs/2601.18517>)

Hong Kong SAR

James Sungarda, HONGKAI LIU, ZILONG ZHOU, TIEN-HSUAN WU, JOHNSON C.S. CHEUNG, BEN KAO

8 Dec. 2025

- Explored novel methods of integrating cognitive models and learning pedagogies with Generative AI as an educational tool in the social work domain, focusing on **high-fidelity client simulations** and **real-time feedback mechanisms**.
- Trained **BERT-based language models** and **LLM few-shot learning** for multi-label classification, enhancing contextual awareness by using domain specific pre-training and achieving **97%** accuracy rate.
- Organized and processed large scale **text annotation** tasks of over **13,000+** entries by collaborating with social work professionals to ensure high-quality labelled datasets for supervised learning.

Projects

Arang (<https://github.com/invalidcode232/arang>)

Typescript, Next.js, PostgreSQL, Azure

Oct. 2024

- Arang is a **spaced repetition** algorithm flashcards app which combines traditional digital flashcards functionality with LLM models using **Retrieval Augmented Generation (RAG)** to intelligently generate flashcards questions and answers based on images, text, and videos.
- **Full-stack web application** written in TypeScript, using the **Next.js framework with PostgreSQL** as backend database. The app connects to Azure's OpenAI service as the LLM provider.

Timeblockr (<https://github.com/invalidcode232/timeblockr>)

Typescript

Jan. 2025 - May. 2025

- Smart AI assisted calendar which uses **RAG** to retrieve and read existing calendar lists e.g. Google Calendar, connects to **Google Maps and OpenWeather** to intelligently assign time slots.
- Fine-tuned LLM generates recommendations and schedule adjustments based on real time data such as **traffic data and weather forecasts**.

Honors & Awards

Sep. 2025 **Third Prize**, iCANX Global Innovation Award 2025.

Hong Kong SAR

Oct. 2024 **Second Runner Up**, HKU CSAxDSA Sociathon 2024.

Hong Kong SAR

Sep. 2024 **Scholar**, HKU Undergraduate Entrance Scholarship for Outstanding Academic Talents.

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Jun. 2022 **Finalist**, National Olympiad in Informatics.

Indonesia